

TECHNICAL PROPOSAL

Consultancy Services: Routine Data Quality Audits (RDQAs)

Tiko Africa – Uganda Programme

Kampala and Wakiso Districts

March – December 2026

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Subject: "Tiko Routine Data Quality Audits - Uganda 2026"

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1. Executive Summary

We are pleased to submit this technical proposal in response to Tiko Africa's call for consultancy services to conduct Routine Data Quality Audits (RDQAs) and facilitate

Strategic Data Review Sessions across Tiko-supported health facilities in Kampala and Wakiso Districts, Uganda.

Our consulting team brings together over 23 combined years of experience in biostatistics, health data quality assurance, monitoring and evaluation, and health management information systems in Uganda. Paul Mubiri, the proposed Lead Consultant, has directly conducted routine data quality assessments across multiple health programmes including the Preterm Birth Initiative (PTBi), Maryland Global Initiative Corporation (MGIC), and the iTECH project at Makerere University, and possesses hands-on proficiency with DHIS2, KoboToolbox, and Uganda's HMIS reporting tools. Raymond R. Wayesu, the proposed Data Analyst and Biostatistician, brings rigorous quantitative skills from the Uganda Virus Research Institute (UVRI) and extensive experience in data quality control, clinical trial data management, and statistical modelling.

Our approach is anchored in three internationally recognised frameworks: the MEASURE Evaluation RDQA Tool, the WHO Data Quality Review (DQR) Toolkit, and Uganda Ministry of Health HMIS guidelines. We propose a dual-system verification methodology that cross-references Tiko Platform data against facility-level HMIS source documents and DHIS2 reported values — bridging Tiko's innovative digital data ecosystem with national reporting standards. This methodology, combined with participatory root cause analysis and structured follow-up mechanisms, will ensure that Tiko's programmatic data is accurate, complete, and actionable.

To demonstrate our technical capabilities and the quality of our analytical outputs, we have developed an interactive RDQA Demo Dashboard showcasing our verification factor analysis, systems assessment visualisations, root cause taxonomy, and data triangulation methodology. The dashboard is accessible at: <https://rdqa-demo.vercel.app/>

We have also developed a full-featured **RDQA Web Tool** — a live R/Shiny application that goes beyond the static demo by providing interactive data entry, real-time Verification Factor computation, M&E systems assessment with radar charts, datimvalidation-style checks, three-way data triangulation, and downloadable reports. The web tool supports both demo mode and live DHIS2 integration. It is accessible at: <https://dnfxdz-raymond-wayesu.shinyapps.io/rdqa-web-tool/>

2. Understanding of the Assignment

Tiko Africa (formerly Triggerise), founded in 2014, is an African non-profit organisation empowering adolescent girls and young women (AGYW) aged 15–24 through improved access to sexual and reproductive health (SRH), HIV, sexual and gender-based violence (SGBV), and mental health services. In Uganda, Tiko operates through partnerships with private franchised and non-franchised facilities, public health facilities, and local government authorities (KCCA and Wakiso District), reaching approximately 54 health service providers, 296 community mobilisers, and 46 local retailers.

The Tiko Platform — accessible via SMS, WhatsApp, Tiko cards, and a light Android app — enables real-time data collection on service delivery, client interactions, and demand generation. While the platform incorporates advanced quality features (AI-based fraud detection, biometric authentication, GPS tracking), a structured and independent RDQA process is essential to systematically verify reported data against primary source documents (HMIS registers) and the national DHIS2 system.

We understand the core challenge: Tiko's digital data pipeline and Uganda's paper-based HMIS reporting system operate in parallel. Discrepancies between these systems — whether caused by transcription errors, staff capacity gaps, stock-outs of HMIS tools, misunderstanding of the Tiko app, or intentional data falsification — must be identified, root-caused, and resolved through collaborative corrective action. The RDQA must also account for broader contextual factors such as donor funding shifts, new health policies, and facility-level resource constraints.

The consultancy demands a consultant team that combines deep technical knowledge of Uganda's HMIS/DHIS2 infrastructure with the ability to work within Tiko's technology-driven environment, coordinate effectively with KCCA and Wakiso District health authorities, and foster a culture of data use and ownership at facility level. Our team meets all of these requirements.

3. Technical Approach and Methodology

3.1 Guiding Frameworks

Our methodology integrates three complementary frameworks to ensure rigour, credibility, and alignment with national standards:

- **MEASURE Evaluation RDQA Tool:** Provides the core structure for data verification (Part 1: recounting from source documents and computing Verification Factors) and systems assessment (Part 2: scoring M&E capacity across five functional areas).
- **WHO Data Quality Review (DQR) Toolkit:** Frames the assessment around four quality dimensions — completeness and timeliness, internal consistency, external consistency, and population data comparisons.
- **Uganda MoH HMIS Guidelines:** Ensures all verification activities align with national data quality attributes (accuracy, completeness, validity, consistency, uniqueness, and timeliness) and use official HMIS registers (HMIS 031, 055B, 071, 074, 080/081) and reporting tools (HMIS 105, HMIS 106a).

3.2 Dual-System Verification Approach

The unique value of this RDQA lies in its three-way data triangulation across:

Data Source	Description	Verification Method
Tiko Platform	Real-time digital records of services delivered, client interactions, and provider validation via the Tiko app/ SMS/WhatsApp	Extract Tiko reports for selected indicators and reporting periods; compare aggregated counts against facility registers
Facility Source Documents	Primary HMIS registers (HMIS 031, 055B, 071, 074, 080/081), tally sheets, and monthly summary reports (HMIS 105)	Physical recounting from registers for selected indicators; compute Verification Factor (VF = Recounted / Reported × 100%)
National DHIS2	District-level aggregate data as entered by District Biostatisticians from facility HMIS 105 reports	Extract DHIS2 data for same indicators/periods; compare against both facility reports and Tiko data

A Verification Factor (VF) within the ±10% tolerance band (90–110%) is considered acceptable. Values below 90% signal potential over-reporting; values above 110% indicate under-reporting. All discrepancies outside this band will trigger root cause investigation.

3.3 Monthly RDQA Process

Each monthly RDQA cycle will follow a structured sequence of activities:

Phase 1: Pre-Visit Preparation (Days 1–3)

- Receive facility list and indicator priorities from Tiko for the audit month.
- Conduct desk review of Tiko Platform data exports and DHIS2 data using the WHO Data Quality App to pre-identify facilities and indicators with potential quality issues.
- Coordinate with KCCA and Wakiso District Health Teams for joint participation in field visits.
- Prepare and pre-load digital RDQA instruments on KoboToolbox with facility-specific cascading selects, indicator definitions, and auto-calculated verification ratios.
- Review status of action points from the previous month's RDQA.

Phase 2: On-Site RDQA Execution (Days 4–18)

- Deploy field audit sub-teams (2–3 persons per facility) for systematic data verification.
- At each facility: (a) introduce the RDQA purpose and process to facility staff; (b) extract and recount data from primary HMIS registers for selected indicators; (c) compare recounted values against Tiko Platform reports and HMIS 105 monthly summaries; (d) compute Verification Factors for each indicator; (e) photograph register pages and reports as evidence; (f) assess the completeness and condition of health records.
- Document contextual observations: staffing levels, register stock-outs, data management practices, facility infrastructure, and any broader factors (donor shifts, policy changes) affecting data quality.

- Conduct brief qualitative interviews with facility in-charges and data personnel for root cause analysis of identified discrepancies, using a structured interview guide exploring staff attitudes, training gaps, tool usability, and systemic barriers.
- Verify closure of action points from previous audit cycles with facility staff, collecting evidence of resolution.
- Provide on-the-spot verbal feedback and agree preliminary corrective actions collaboratively with facility staff.

Phase 3: Analysis and Reporting (Days 19–25)

- Clean and validate field data from KoboToolbox exports.
- Compute final Verification Factors by indicator and facility; generate facility-level scorecards and systems assessment spider charts.
- Compile Individual Facility Reports documenting variances, root causes, and recommended corrective actions.
- Update the Action Point Tracker with status classifications (“Closed,” “Pending,” or “Overdue”) and supporting evidence.
- Submit Final Cleaned Dataset and Comparative Checks Report to Tiko by the 25th of the month.

3.4 Quarterly Facility Data Review Sessions

Quarterly, we will support Tiko in executing Facility Data Review Sessions that go beyond compliance to drive continuous performance improvement. Each session will:

- Bring together facility in-charges, providers, data personnel, mobilisers (YAPs), and Tiko staff in a participatory forum.
- Present before/after performance trend analysis covering the previous six months, using visualisations of Verification Factors, service delivery volumes, and demand generation metrics.
- Present Quality Assurance (QA) findings and explicitly link data discrepancies to the Facility Improvement Plan (FIP), highlighting financial and reputational implications.
- Facilitate collaborative discussion to identify systemic patterns, agree strategic goals, and generate commitments for the next quarter.
- Document attendees, performance trends discussed, and specific commitments in the Facility Data Review Report.

3.5 Quarterly Systemic Analysis

At the end of each quarter, we will produce a Strategic Insights Report that aggregates findings across all audited facilities to identify cross-division patterns (Kampala vs. Wakiso), systemic gaps requiring high-level Tiko intervention, and evidence-based recommendations for programme-wide data quality strengthening.

3.6 Root Cause Analysis Framework

Discrepancies will be investigated using a structured root cause taxonomy:

Category	Examples	Investigation Approach
Human Factors	Staff attitude, training gaps, high turnover, competing clinical priorities	Staff interviews, observation of data recording practices, training needs assessment
Systems/ Process	Transcription errors, misunderstanding of Tiko app, unclear indicator definitions	Process mapping, error pattern analysis, app usability assessment
Supply/ Resources	Stock-outs of HMIS registers/forms, lack of data storage, connectivity issues	Physical inspection, supply chain review, infrastructure assessment
Intentional	Data falsification, fabrication, selective reporting	Cross-verification with multiple sources, patient-level spot checks, pattern analysis
Contextual/ External	Donor funding shifts, new MoH policies, disease outbreaks, political changes	Policy review, stakeholder interviews, trend analysis against external events

3.7 Digital Data Collection Tools

We will use KoboToolbox as the primary digital data collection platform for on-site RDQA, leveraging its offline capability (critical for facilities with unreliable connectivity), skip logic, real-time verification ratio auto-calculation, GPS auto-capture, and photo documentation features. We will develop a custom KoboToolbox RDQA instrument adapted to Uganda's HMIS registers with four modules: Facility Identification, Data Verification (with auto-calculated VFs), Systems Assessment (Likert-scale scoring across five functional areas), and Action Planning. DHIS2 desk review will utilise the WHO Data Quality App already available in Uganda's national DHIS2 instance. We will work with the Tiko technology team to align our data collection with Tiko's existing platform and reporting systems.

In addition to KoboToolbox, we have developed an interactive RDQA Demo Dashboard that demonstrates our analytical capabilities, including Verification Factor visualisation, systems assessment spider charts, root cause taxonomy analysis, monthly process timelines, and three-way data triangulation flows. This dashboard illustrates the type of outputs and analytical rigour we will bring to the Tiko engagement. Interactive Demo: <https://rdqa-demo.vercel.app/>

Beyond the demo dashboard, we have built a fully functional **Routine Data Quality Audit Web Tool (RDQA Web Tool)** using R/Shiny. While the demo dashboard (rdqa-demo.vercel.app) is a static Next.js website showcasing our analytical methodology and sample outputs, the RDQA Web Tool is a live, interactive application where users can load data (demo or live DHIS2),

compute Verification Factors in real time, run datimvalidation-style checks, perform M&E systems assessments with spider charts, and generate downloadable HTML and CSV reports.

RDQA Web Tool: <https://dnfxdz-raymond-wayesu.shinyapps.io/rdqa-web-tool/>

3.8 Coordination and Engagement Strategy

Effective coordination with local government authorities is essential to the success and credibility of the RDQA process. Our engagement strategy includes:

- Formal introduction meetings with KCCA Health Department and Wakiso District Health Officer at inception to present the RDQA scope and seek joint participation.
- Invitation of District Biostatisticians and Health Information Officers to participate in facility visits, strengthening government ownership and facilitating DHIS2 data access.
- Monthly coordination calls with Tiko programme team to align facility schedules, share preliminary findings, and adjust audit plans as needed.
- Sharing of summary findings with district health teams (with Tiko's prior written consent) to strengthen the data quality feedback loop.

4. Work Plan and Timeline

Activity	Time Frame	Deliverable	Responsible
Project Inception	1–13 Mar 2026	Inception Report (.ppt): team structure, work plan, risk matrix, KCCA/Wakiso engagement strategy	Lead Consultant (PM)
Tool Development & Pilot	10–17 Mar 2026	Custom KoboToolbox RDQA instrument; pilot report from 2–3 facilities	Data Analyst (RW) + Lead
Monthly RDQA Cycle 1	1–25 Mar 2026	Cleaned dataset, facility reports, action point tracker	Full team
Monthly RDQAs	By 25th monthly (Apr–Dec)	Monthly cleaned datasets, individual facility reports, action point tracker	Full team
Post-RDQA Follow-ups	1st week of each month	Follow-up verification of previous action points	Field auditors + Lead
Q1 Data Review Session	Last week of Jun 2026	Facility Data Review Report (Q1)	Lead Consultant + Tiko
Q2 Data Review Session	Last week of Sep 2026	Facility Data Review Report (Q2)	Lead Consultant + Tiko
Q3 Data Review Session	Last week of Dec 2026	Facility Data Review Report (Q3) + Final report	Lead Consultant + Tiko
Quarterly Systemic Analysis	Jun, Sep, Dec 2026	Strategic Insights Report (Kampala vs. Wakiso patterns, systemic recommendations)	Lead + Data Analyst

Monthly cycle summary: Days 1–3 (preparation and desk review) → Days 4–18 (on-site facility visits and data verification) → Days 19–25 (analysis, reporting, and submission). The exact number of facilities per month will be communicated by Tiko in advance.

5. Team Composition and Management

Our core team comprises two senior professionals with complementary expertise, supplemented by field auditors who will be mobilised based on the number of facilities per audit cycle.

Role	Name	Key Qualifications	RDQA Responsibilities
Lead Consultant / Team Leader	Paul Mubiri	MSc Biostatistics (Makerere); 13+ years in health M&E, DQA, and data management; hands-on RDQA experience with PTBi, MGIC, and iTECH; proficient in DHIS2, KoboToolbox, STATA, R, REDCap; published researcher with 20+ peer-reviewed publications	Overall technical leadership; RDQA tool design; KCCA/ Wakiso coordination; root cause analysis; quarterly data review facilitation; report authorship; quality assurance
Data Analyst / Biostatistician	Raymond R. Wayesu	MSc Statistics (Linköping University); 10+ years in biostatistics, data science, and clinical trials; data quality and integrity experience at UVRI, Sanyu Africa, NSSF; proficient in R, Python, SQL; GCP certified; published researcher	Digital RDQA instrument development; Verification Factor computation; DHIS2 data extraction and analysis; dashboard and scorecard generation; systems assessment analysis; statistical reporting
Field Data Quality Auditors (2–4)	To be recruited	Bachelor's degree in public health, statistics, or related field; 3+ years HMIS experience; familiarity with Uganda MoH registers; based in Kampala/ Wakiso	On-site register recounting; data extraction and tallying; photo documentation; preliminary root cause interviews; action point follow-up verification

Team management: Paul Mubiri will serve as the single point of contact with Tiko's programme team. Weekly internal team briefings will ensure consistent tool application and data quality. Raymond Wayesu will provide centralised data management and analysis support, enabling rapid turnaround of facility reports. Field auditors will be supervised by the Lead Consultant and will operate in sub-teams of 2–3 persons per facility, enabling coverage of 3–4 facilities per day.

5.1 Team Gaps and Recruitment Plan

We will recruit 2–4 Field Data Quality Auditors with specific experience in HMIS register review and facility-based data verification in Kampala and Wakiso. Recruitment criteria include a bachelor's degree in public health, statistics, or a related field; at least 3 years of experience working with Uganda's HMIS tools; and demonstrated familiarity with digital data collection platforms. Recruitment will be completed during the inception

phase (by 13 March 2026) and all field auditors will undergo a 2-day training and pilot exercise before the first full RDQA cycle.

6. Quality Assurance Plan

Data quality is the product we are assessing — and the standard to which we hold our own work. Our quality assurance plan includes:

- **Standardised tools:** All field auditors will use identical KoboToolbox instruments with built-in validation rules (integer constraints, skip logic, mandatory fields) to minimise data entry errors.
- **Inter-rater reliability:** During the pilot phase, two auditors will independently assess the same facility to calibrate consistency. Any discrepancies exceeding 5% will trigger re-training.
- **Photographic evidence:** All register recounts will be supported by photographs of the relevant register pages, forced-capture (no gallery selection) via KoboCollect, providing an auditable evidence trail.
- **Audit metadata logging:** KoboToolbox audit logging will track form start/end times, GPS coordinates, and enumerator identity, enabling supervision of field team performance.
- **Same-day data review:** The Data Analyst will review submitted KoboToolbox records daily during fieldwork, flagging incomplete or inconsistent entries for same-day correction.
- **Independent verification:** The Lead Consultant will conduct spot-check re-verification at a random 10% of audited facilities each month.

7. Deliverables

Deliverable	Description	Format	Frequency
Inception Report	Team structure, work plan, risk matrix, KCCA/Wakiso engagement strategy, adapted RDQA tools	PowerPoint slide deck + KoboToolbox XLSForm	Once (by 13 Mar)
Final Cleaned Dataset	Complete verified data using Tiko RDQA tools	Excel/CSV (Tiko format)	Monthly (by 25th)
Individual Facility Reports	Variances, root causes, and recommended corrective actions per facility	Word/PDF with VF tables and spider charts	Monthly (by 25th)
Action Point Tracker	Status of all action points: Closed, Pending, or Overdue with evidence	Excel tracker with evidence links	Monthly (by 25th)
Facility Data Review Report	Summary of deep-dive sessions: attendees, trends, commitments	Word/PDF + presentation slides	Quarterly
Strategic Insights Report	Cross-division patterns (Kampala vs. Wakiso), systemic gaps, recommendations	Word/PDF with data visualisations	Quarterly

8. Past Performance and References

Our team members have delivered data quality assessments and health data management services for the following organisations. References are available upon request:

- **Preterm Birth Initiative (PTBi), Makerere University School of Public Health (2016–2019):** Paul Mubiri served as Biostatistician conducting routine data quality assessments, data quality assurance, and health facility assessments across implementation science study sites in Eastern Uganda. This included direct register verification, data coding, and stakeholder presentation of findings.
- **Maryland Global Initiative Corporation / University of Maryland Baltimore (2022–2023):** Paul Mubiri supported project data analysis, data quality assessment and improvement, tool design, Power BI visualisation, and capacity building with the Strategic Information and M&E team under the PEPFAR consortium.
- **Makerere University Hospital – iTECH Project (2021–2024):** Both Paul Mubiri and Raymond Wayesu contributed to data quality assessment, statistical analysis, and prediction modelling for this maternal health research project, with Paul leading protocol and SOP development.

- **Sanyu Africa Research Institute – BabyGel Study (2023–2024):** Raymond Wayesu served as Hub Data Supervisor for a cluster-controlled trial across 72 villages in Eastern Uganda, ensuring quality data collection by hub data collectors and accurate entry into electronic data capture systems.
- **Uganda Virus Research Institute – CEPI Project (2024–present):** Raymond Wayesu serves as Biostatistician applying advanced statistical methods to immunology studies, with responsibility for research design, data interpretation, and quality assurance of laboratory data.

Combined publication record: The team has co-authored over 25 peer-reviewed publications in leading journals including The Lancet Global Health, BMC Health Services Research, BMC Public Health, PLOS ONE, and Human Resources for Health.

9. Budget Framework

A detailed budget breakdown will be provided in a separate financial proposal. The following framework outlines the major cost categories and assumptions:

Cost Category	Description	Basis of Estimation
Professional Fees	Lead Consultant, Data Analyst, Field Auditors	Daily rates per role; number of working days per month
Transport and Logistics	Vehicle hire, fuel, field team travel within Kampala and Wakiso	Per facility visit; estimated distances and facility density
Data Collection Tools	KoboToolbox (free tier), printing of backup forms, mobile data	Lump sum per month
Communication	Airtime, internet, coordination calls	Monthly lump sum
Report Production	Printing, binding, data visualisation software	Per deliverable
Training and Piloting	Field auditor training (2 days), pilot at 2–3 facilities	One-time during inception
Quarterly Review Sessions	Venue (if needed), materials, facilitation	3 sessions over the assignment period
Contingency (7%)	Unforeseen costs, additional facilities, extended fieldwork	Percentage of subtotal

Note: Final costing will be calibrated to the confirmed number of facilities per month, as communicated by Tiko in advance of each audit cycle. The budget is structured to offer value for money by leveraging existing digital tools (KoboToolbox free tier, DHIS2 WHO DQ App), deploying locally-based team members (eliminating accommodation costs), and using efficient parallel sub-team deployment to maximise facility coverage per field day.

End of Technical Proposal

Interactive RDQA Demo: <https://rdqa-demo.vercel.app/>

RDQA Web Tool (R/Shiny): <https://dnfxdz-raymond-wayesu.shinyapps.io/rdqa-web-tool/>

Paul Mubiri & Raymond R. Wayesu | February 2026